

Student Record Management System Report

Student Information

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Internship: C++ Programming Internship

Project Title: Student Record Management System

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1. Objective

The objective of this project is to develop a console-based Student Record Management System using C++. The project helps in understanding the fundamental concepts of C++ programming such as variables, loops, functions, arrays, structures, and file handling. It also improves problem-solving skills by implementing a menu-driven application.

2. Introduction

The Student Record Management System is a simple application that allows users to manage student records efficiently. Users can add, view, search, update, and delete student information. The program also saves records in a file so that data is available even after the program is closed and reopened.

3. Concepts Used

- Variables
- Data Types
- Functions
- Arrays
- Structures ("struct")
- Loops ("for", "while", "do-while")
- Conditional Statements ("if", "switch")
- File Handling ("ifstream", "ofstream")
- Menu-Driven Programming
- Input Validation

4. Features

The system provides the following features:

- Add Student Record
- View All Student Records
- Search Student by Roll Number
- Update Student Information
- Delete Student Record
- Save Records into a File
- Load Records from a File
- Input Validation

- Menu-Driven Interface

5. Program Description

Add Student

This function allows the user to enter a student's roll number, name, age, and marks. The information is stored in an array of structures.

View Students

This function displays all stored student records along with their details.

Search Student

The program searches for a student using the roll number entered by the user. If found, the student's information is displayed.

Update Student

The user enters a roll number, and if the student exists, the program allows updating the student's name, age, and marks.

Delete Student

The program deletes a student's record using the roll number and shifts the remaining records to remove the empty space.

Save Records

This function stores all student records in a text file using file handling.

Load Records

Whenever the program starts, this function reads the saved records from the file and loads them into the program.

6. Input Validation

The program validates user input to improve reliability.

- Menu choice must be between 1 and 7.
- Roll number must be positive.
- Age must be within a valid range.
- Marks must be between 0 and 100.
- Duplicate roll numbers are not allowed.
- Student name cannot be empty.

7. Sample Output

(Paste screenshots of your program here.)

- Main Menu

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student
- Save Records

8. Conclusion

This project helped me understand the basic concepts of C++ programming, including structures, arrays, functions, loops, and file handling. It also improved my logical thinking and problem-solving skills. Through this project, I learned how to design a menu-driven application and manage data efficiently using file storage.

9. Future Improvements

In the future, this project can be enhanced by:

- Adding a graphical user interface (GUI)
- Connecting the program with a database
- Implementing password-based login
- Adding sorting and filtering options
- Generating student reports automatically

10. References

- C++ Programming Language Notes
- Devixo Solutions Internship Task
- C++ File Handling Documentation

11. Output

STUDENT RECORD MANAGEMENT SYSTEM

1. Add Student
2. View Student
3. Search Student
4. Update Student
5. Delete Student
6. Save Records
7. Exit

Enter your choice: 1

Enter your roll number: 4

Enter your name: iqbal

Enter your age: 29

Enter your marks: 78

student added successfully.

STUDENT RECORD MANAGEMENT SYSTEM

1. Add Student
2. View Student
3. Search Student
4. Update Student
5. Delete Student
6. Save Records
7. Exit

Enter your choice: 2

Student 1

Roll number: 4

Name: iqbal

Age: 29

Marks: 78

STUDENT RECORD MANAGEMENT SYSTEM

1. Add Student
2. View Student
3. Search Student
4. Update Student
5. Delete Student
6. Save Records
7. Exit

Enter your choice: 3

Enter the roll number to search: 4

Student found

Roll number: 4

Name: iqbal

Age: 29

Marks: 78

STUDENT RECORD MANAGEMENT SYSTEM

1. Add Student
2. View Student
3. Search Student
4. Update Student
5. Delete Student
6. Save Records
7. Exit

Enter your choice: 6

Records saved successfully.

Records saved successfully.

STUDENT RECORD MANAGEMENT SYSTEM

1. Add Student
2. View Student
3. Search Student
4. Update Student
5. Delete Student
6. Save Records
7. Exit

Enter your choice: 7

Thank You!Exiting program.

Process returned 0 (0x0) execution time : 70.268 s

Press any key to continue.